ASSIGNMEN 7-1

Fundamentals of Cyber Security (TCS-495)

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Git Link -> priyanshi-r76/Cyber Security Assignment/

Ans:1 Importance of Cyber Security

Cyber Security protects our personal data, money, privacy, and digital system from hacking, fraud and missuse. It is essential for safe online activities and smooth functioning of businesses and services.

Real Life Example.

1. Online payments - Apps like Google Pay, & phone Pay need security to prevent fraud.
2. Social media - Platform like Instagram protects personal accounts from hacking.
3. Online Shopping - Sites like Amazon secure card and address details
4. Healthcare - Protects sensitive patient records in hospitals
5. Education platform - Apps like Google Classroom keep student data safe.

Cyber security keeps our digital life safe & secure.

Ans. 2.

Type	Intent	Legality	Activity Description.
White hat	Good (ethical)	Legal	Finds and fixes security vulnerability.
Grey Hat	Mixed (no harmful intent)	Semi-Legal	Explores systems without permission but may report issues.
Black Hat	Malicious	Illegal	Hacks system to steal data or cause harm/danger.

Real World Scenarios:

- White Hat - security expert working with companies like Google to find bugs and improve system security.
- Grey Hat - A hacker discovers a website flaws with permission and informs the company instead of exploring it.
- Black Hat - Hackers involved in the WannaCry ransomware attack encrypting data and demanding ransom.

Ans. 3 Zero Trust Model

The Zero Trust Model is a security approch based on "never trust, always verify"

No user or device is trusted automatically - even if they are inside the network. Every access request must be verified using identity, device state, and permission.

~~Secu~~ Security Strategy for a Computer Lab :-

1. User Authentication.
   - every student log in with student ID + password
   - multi factor authentication.
2. Device Verification
   - Only authorized lab computer can access the network
   - Block unknown USB devices & external device
3. Least Privilage Access.
   - Student get only required access
   - Teachers/admins have higher permission.
4. Network Segmentation.
   - Separate networks for student, faculty & admin systems.
   - Sensitive data stored in restricted segments.
5. ~~Conti~~ Continous Monitering
   - Track login activity and ~~unusu~~ unusual behaviour
   - Auto logout inactive users.
6. Data Protection.
   - Encrypts imp data
   - Regular backups of lab system
7. Regular Update & Patching
   - keep OS & software updated to avoid vulnerabilities.

Ans.4- CIA Triad - The core principales of Cyber Security.

1. Confidentiality
   - ensures that sensitive data is accessed only by authorized users.
   - Cyberattack Example: Data breach like Facebook- Cambridge Analytica data scandal where user data was exposed.

2. Integrity
   - ensure data is accurate & not altered w/o permission.
   - Cyberattack Example: Hackers modifying website data or databases (eg: changing student marks in a college system)

3. Avaliability
   - Ensures system and data are ~~are~~ accessible whenever needed.
   - Cyber attack Ex:- DDOS attack on Dyn which made many websites unavailable.

Ans.5.

a) Possible Threats
   - Account compromise (stolen credentials)
   - Unauthorized access
   - Log tampering

b) Analyse first
   - Login details (IP, time)
   - Affected accounts
   - Modified files & available logs.

c) Treatement Steps.
   - Lock accounts & reset passwords.
   - Enable MFA
   - Isolate system
   - Restore data from backups
   - Scan for malware.

d) CIA Affected.
   - Confidentiality → unauthorised access
   - Integrity → data modified
   - Avaliability → system reliability affected.

Ans. 6.

a) Ransomware attack (files got encrypted)

b) 1. Cracked software contained malware.
2. On installation, it executed and encrypted

c) 1. Avoid private/cracked software
2. Use trusted antivirus
3. Keep system updated
4. Download only from official sources

d) 1. Restore files from backup
2. Use antivirus to remove malware
3. Avoid paying ransom

Q1-12 → completed → github

Bonus question → completed → github.